

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions

Remote

- Built a resilient **Puppeteer/Node.js backend automation workflow** for MCA filings with retries, session recovery, public API lookups, state validation, and human-reviewable generated drafts, reducing manual filing time by **85%**.
- Integrated **browser automation** with structured backend state, CAPTCHA handoff, network-idle monitoring, and recoverable checkpoints for unreliable government-site workflows.
- Containerized services with **Docker**, deployed to **Google Cloud Run**, and automated releases through **GitHub Actions**, improving deployment velocity by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built Python medical-image processing/inference tooling for **95 brain subparts**, handling preprocessing, orientation correction, model execution, generated masks, and large-volume MRI data paths.
- Reduced runtime by **87%** with **async/parallel execution** and debugged pipeline issues across conformation, orientation, label mapping, and PyTorch/ONNX-style inference.
- Worked inside a mature open-source codebase, adapting UI/data-path changes to existing **VTK, wxPython, NumPy**, and model-inference boundaries.

Open Source

Contributions 

- Contributed patches across **Invesalious, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant**, working through maintainer review on reliability, security, and integration-quality fixes.
- Worked across unfamiliar **systems/cloud-native codebases**, reading existing architecture and shipping focused fixes without breaking project conventions or maintainer expectations.

Projects

Swish Support System | Python, FastAPI, PostgreSQL, Redis, Langfuse, Docker

GitHub 

- Built **Python/FastAPI backend services** for support automation with **PostgreSQL/Redis** state, session recovery, case lifecycles, action records, and deterministic refund/replacement/escalation policy.
- Implemented backend reliability tooling around the agent using **Langfuse/local JSON traces**, multi-turn replay, deterministic validators, optional judge scoring, JSONL outputs, and regression cases for failed behaviors.
- Added safeguards around **rate limits, timeouts, session clearing, case-status verification**, policy-safe response checks, and fail-on-error suite execution for CI-style regression testing.

Murdoc | Python, FastAPI, MCP, OpenTelemetry, Prometheus, Docker

GitHub 

- Instrumented a gateway service with **OpenTelemetry traces, Prometheus metrics**, route profiles, health checks, and audit records tying model/tool requests to policy outcomes.
- Built **replayable decision records** across prompt, tool, retrieval, and output layers, keeping model/tool policy decisions inspectable without forcing downstream services to change execution paths.

Postificus | Go, Echo, Rod, PostgreSQL, Redis, RabbitMQ, Docker, Prometheus

GitHub 

- Built Go/Echo backend services for a content distribution platform with REST APIs, **RabbitMQ workers**, PostgreSQL persistence, Redis autosave, and recoverable job state.
- Added **retries, DLQs, circuit breakers, health checks**, and Prometheus metrics so unreliable external API/browser workflows stayed observable and recoverable.

Seaweed Contest Platform | Go, PostgreSQL, Redis, Judge0, AWS, Docker, Kubernetes

GitHub 

- Built backend services for registration, code submissions, **sandboxed Judge0 execution**, scoring, and live rankings across **500+ concurrent users**; reached **sub-5s P95 verdict latency**.
- Designed **async verdict polling**, submission state transitions, PostgreSQL result schemas, Redis-backed runtime state, and **Docker/Kubernetes** deployment on AWS for contest reliability.

Technical Skills

Languages: Go, Python, TypeScript, C++

Frameworks: FastAPI, Echo, Node.js, Express, PyTorch

Tools: PostgreSQL, Redis, RabbitMQ, Docker, Kubernetes, AWS, GCP, GitHub Actions, OpenTelemetry, Prometheus, Langfuse, Judge0